

Generalizations and Enhancements to FLRW and Friedmann Equations

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I: An FLRW Generalization

I start by developing a generalization of the FLRW equations. As I take steps to reduce it to the FLRW solution, the discrepancy appears.

The initial Lambda vacuum field equations are:

$$\begin{aligned}
 R_{00}: \quad & \frac{3\ddot{a}}{aA} - \frac{3\dot{a}\dot{A}}{2aA^2} - \frac{A'}{a^2Ar_B} + \frac{(A')^2}{4a^2A^2B} - \frac{A'C'}{2a^2ACB} + \frac{A'B'}{4a^2AB^2} - \frac{A''}{2a^2AB} = \Lambda \\
 R_{01}: \quad & -\frac{\dot{a}A'}{aA} = 0 \\
 R_{11}: \quad & \frac{2\dot{a}^2}{a^2A} + \frac{\ddot{a}}{aA} - \frac{\dot{a}\dot{A}}{2aA^2} + \frac{(A')^2}{4a^2A^2B} - \frac{2C'}{a^2r_{CB}} + \\
 & \frac{(C')^2}{2a^2C^2B} + \frac{B'}{a^2r_B^2} + \frac{A'B'}{4a^2AB^2} + \frac{C'B'}{2a^2CB^2} - \frac{A''}{2a^2AB} - \frac{C''}{a^2CB} = \Lambda \\
 R_{22}: \quad & \frac{1}{a^2r_C^2} - \frac{1}{a^2r_B^2} + \frac{2\dot{a}^2}{a^2A} + \frac{\ddot{a}}{aA} - \frac{\dot{a}\dot{A}}{2aA^2} - \\
 & \frac{A'}{2a^2Ar_B} - \frac{2C'}{a^2r_{CB}} - \frac{A'C'}{4a^2ACB} + \frac{B'}{2a^2r_B^2} + \frac{C'B'}{4a^2CB^2} - \frac{C''}{2a^2CB} = \Lambda
 \end{aligned}$$

Since $a(t)$ is assumed not to be constant, from R_{01} we get that

$$A(t, r) = \mathcal{A}[t] \quad \text{where } \mathcal{A}(t) \text{ is a new unknown.}$$

R_{01} becomes an identity, $0=0$, and we

are left with 3 equations in the 3 unknowns $\mathcal{A}$, B , and C :

$$\begin{aligned}
 R_{00}: \quad & \frac{3\ddot{a}}{a\mathcal{A}} - \frac{3\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2} = \Lambda \\
 R_{11}: \quad & \frac{2\dot{a}^2}{a^2\mathcal{A}} + \frac{\ddot{a}}{a\mathcal{A}} - \frac{\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2} - \frac{2C'}{a^2r_{CB}} + \frac{(C')^2}{2a^2C^2B} + \frac{B'}{a^2r_B^2} + \frac{C'B'}{2a^2CB^2} - \frac{C''}{a^2CB} = \Lambda \\
 R_{22}: \quad & \frac{1}{a^2r_C^2} - \frac{1}{a^2r_B^2} + \frac{2\dot{a}^2}{a^2\mathcal{A}} + \frac{\ddot{a}}{a\mathcal{A}} - \frac{\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2} - \frac{2C'}{a^2r_{CB}} + \frac{B'}{2a^2r_B^2} + \frac{C'B'}{4a^2CB^2} - \frac{C''}{2a^2CB} = \Lambda
 \end{aligned}$$

Solving R_{00} for $\mathcal{A}$ yields

$$(i) \quad \mathcal{A}(t) = \frac{3\dot{a}(t)^2}{\Lambda a(t)^2 + 3|k|} \quad \text{where } k \in \mathbb{R} \text{ is a constant of integration.}$$

This is where the absolute value first appears.

To generalize the FLRW equations, this would have to have been

$$(ii) \mathcal{A}(t) = \frac{3 \dot{a}(t)^2}{\Lambda a(t)^2 - 3k}.$$

In fact, (ii) is also a legitimate solution. However, another solution is

$$(iii) \mathcal{A}(t) = \frac{3 \dot{a}(t)^2}{\Lambda a(t)^2 + 3k}.$$

(ii) makes $\mathcal{A}(t) > 0$ for all $k < 0$ and (iii)

makes $\mathcal{A}(t) > 0$ for all $k > 0$. (i) makes $\mathcal{A}(t) > 0$ for all k .

Plugging in $\mathcal{A}(t)$ from (i), we are left with 2 equations in the 2 unknowns, B and C :

$$\begin{aligned} R_{11}: \quad & \frac{2|k|}{a^2} + \frac{B'}{a^2 B^2 r} - \frac{2C'}{a^2 B r C} - \frac{C''}{a^2 B C} + \frac{B' C'}{2 a^2 B^2 C} + \frac{(C')^2}{2 a^2 B C^2} + \Lambda = \Lambda \\ R_{22}: \quad & \frac{2|k|}{a^2} - \frac{1}{a^2 B r^2} + \frac{1}{a^2 r^2 C} + \frac{B'}{2 a^2 B^2 r} - \frac{2C'}{a^2 B r C} - \frac{C''}{2 a^2 B C} + \frac{B' C'}{4 a^2 B^2 C} + \Lambda = \Lambda \end{aligned}$$

We can solve for $B(r)$ algebraically:

$$\begin{aligned} R_{11} - 2R_{22}: \quad & -\frac{2|k|}{a^2} + \frac{2}{a^2 B r^2} - \frac{2}{a^2 r^2 C} + \frac{2C'}{a^2 B r C} + \frac{(C')^2}{2 a^2 B C^2} - \Lambda = -\Lambda \\ B(r) = \quad & \frac{(r C'(r) + 2 C(r))^2}{4 C(r) (r^2 |k| C(r) + 1)} \end{aligned}$$

k is the curvature constant. k can be any real number, positive, zero, or negative.

Plugging this in makes the 2 remaining equations into identities:

$$R_{11}: \quad \Lambda = \Lambda$$

$$R_{22}: \quad \Lambda = \Lambda$$

All of the equations are satisfied and so we have an exact solution to Λ field equations:

$$A(t, r) = \mathcal{A}[t] = \frac{3 \dot{a}(t)^2}{\Lambda a(t)^2 + 3|k|}, \quad \text{where } k \text{ is a free constant,}$$

$$B(t, r) = a[t]^2 B(r) = \frac{a(t)^2 (r C'(r) + 2 C(r))^2}{4 C(r) (r^2 |k| C(r) + 1)},$$

$$C(t, r) = a[t]^2 C(r), \quad \text{where } C(r) \text{ is a free parameter,}$$

and line element

$$ds^2 = \frac{3 \dot{a}(t)^2}{\Lambda a(t)^2 + 3 |k|} dt^2 - \frac{a(t)^2 (r C'(r) + 2 C(r))^2}{4 C(r) (r^2 |k| C(r) + 1)} dr^2 - r^2 a(t)^2 C(r) d\theta^2 - r^2 a(t)^2 \sin^2(\theta) C(r) d\phi^2$$

As we will see below in III, the FLRW equations are a special case of this solution.

I refer to this solution as the **Generalized FLRW metric**. It also is a template for identifying the form of higher dimensional FLRW metrics.

II: Set $g_{00}=c^2$ and solve for $a(t)$

The scale factor $a(t)^2$ represents a scaling of proper distance (i.e., of spacetime) and, so, is always positive. Since the scale factor is always represented by $a(t)^2$, negative $a(t)$ is indistinguishable from positive $a(t)$. Importantly, $a(t)$ cannot be imaginary. However, the slope, $\dot{a}(t)$, can be either positive or negative and represents the expanding or contracting of spacetime. Similarly, $\ddot{a}(t)$ can be either positive or negative and represents an acceleration or deceleration of expansion or contraction.

Because $a(t)$ and $-a(t)$ are indistinguishable, even though many of the equations below for $a(t)$ yield both positive and negative solutions, I discard the negative formulas and work with just the positive ones.

The 4D FLRW equation has $g_{00} = A(t, r) = \mathcal{A}(t) = c^2$. With the

aim of deriving the FLRW solution, I solve $\frac{3 \dot{a}^2}{a^2 \Lambda + 3 |k|} = c^2$ for $a(t)$:

$$a(t) = \frac{\sqrt{3} \sqrt{|k|} \sinh\left(\frac{\sqrt{\Lambda} (c t + \nu)}{\sqrt{3}}\right)}{\sqrt{\Lambda}} \text{ where } \nu \text{ is a constant of integration, an offset}$$

$$(1) \quad a(t) = \frac{\sqrt{3} \sqrt{|k|} \sinh\left(\frac{c \sqrt{\Lambda} t}{\sqrt{3}}\right)}{\sqrt{\Lambda}} \text{ where the offset is set to } 0.$$

Formula (1) can be valid when $\Lambda = 0$:

$$\frac{1}{\sqrt{\Lambda}} \sinh\left[\sqrt{\frac{\Lambda}{3}} c t\right] = \lim_{\Lambda \rightarrow 0} \frac{\sinh\left[\sqrt{\frac{\Lambda}{3}} c t\right]}{\sqrt{\Lambda}} = \frac{c t}{\sqrt{3}} \text{ due to L'Hospital's Rule}$$

which implies

$$(2) \quad a(t) = c t \sqrt{|k|}$$

Note that $k \neq 0$ in equations (1) and (2) or else $a(t)=0$, which is not allowed.

However, we can obtain a third solution to $g_{00} = c^2$, one having $k = 0$ (flat space), if we set $k = 0$ before solving the ODE. That is, we solve $\frac{3\dot{a}^2}{a^2\Lambda} = c^2$:

$$(3) \quad a(t) = e^{\pm \frac{ct\sqrt{\Lambda}}{\sqrt{3}}} \alpha \quad \text{where } \alpha \text{ is a constant of integration.}$$

Observe that $\Lambda \geq 0$ and $\alpha > 0$ in order for $a(t)$ to be positive.

Next, observe that $\Lambda \neq 0$ or else $\frac{3\dot{a}^2}{a^2\Lambda}$ is undefined.

Thus, the generalized FLRW solution (3) requires Λ to be positive. I don't see a way to extend (3) to a formula that permits $\Lambda \leq 0$. Even allowing $\alpha \in \mathbb{C}$ does not help.

Combining the solutions, we have:

$$a(t) = \begin{cases} \frac{\sqrt{3} \sqrt{|k|} \sinh\left(\frac{c\sqrt{\Lambda}t}{\sqrt{3}}\right)}{\sqrt{\Lambda}} & \text{if } k \neq 0 \text{ \& } \Lambda \neq 0 \\ ct \sqrt{|k|} & \text{if } k \neq 0 \text{ \& } \Lambda = 0 \\ \alpha e^{\pm \frac{c\sqrt{\Lambda}t}{\sqrt{3}}} & \text{if } k = 0 \text{ \& } \Lambda > 0, \text{ where } \alpha > 0 \text{ is a free constant} \end{cases}$$

III: The FLRW Metric, a subcase of Case I

Plugging in $c(r)=1$ into the Case I solution produces the following subcase:

$$A(t,r) = \mathcal{A}[t] = \frac{3\dot{a}(t)^2}{\Lambda a(t)^2 + 3|k|} \quad \text{where } k \text{ is a free constant,}$$

$$B(t,r) = \frac{a(t)^2}{r^2|k| + 1},$$

$$C(t,r) = a[t]^2,$$

and line element

$$ds^2 = \frac{3\dot{a}(t)^2}{\Lambda a(t)^2 + 3|k|} dt^2 - \frac{a(t)^2}{r^2|k| + 1} dr^2 - r^2 a(t)^2 d\theta^2 - r^2 a(t)^2 \sin^2(\theta) d\phi^2$$

This subcase still generalizes the FLRW metric, which has $A(t,r)$

$= c^2$ and includes the formula for $a(t)$. That is, the FLRW metric is:

$$a(t) = \begin{cases} \frac{\sqrt{3} \sqrt{|k|} \sinh\left(\frac{c\sqrt{\Lambda}t}{\sqrt{3}}\right)}{\sqrt{\Lambda}} & \text{if } k \neq 0 \text{ \& } \Lambda \neq 0 \\ ct \sqrt{|k|} & \text{if } k \neq 0 \text{ \& } \Lambda = 0 \\ \alpha e^{\pm \frac{c\sqrt{\Lambda}t}{\sqrt{3}}} & \text{if } k = 0 \text{ \& } \Lambda > 0, \text{ where } \alpha > 0 \text{ is a free constant} \end{cases}$$

$$A(t,r) = c^2,$$

$$B(t,r) = \frac{a(t)^2}{r^2|k| + 1},$$

$$C(t,r) = a[t]^2,$$

and line element

$$ds^2 = \frac{3 \dot{a}(t)^2}{\Lambda a(t)^2 + 3 |k|} dt^2 - \frac{a(t)^2}{r^2 |k| + 1} dr^2 - r^2 a(t)^2 d\theta^2 - r^2 a(t)^2 \sin^2(\theta) d\phi^2$$